

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 6

End-to-End Study of RNN, LSTM and GRU for Sequence Learning and Video Understanding

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

Lab 6 on GitHub

1. Objective

The objective of this experiment is to develop an end-to-end understanding of recurrent sequence learning by implementing and comparing Vanilla RNN, LSTM and GRU models. The experiment also introduces Backpropagation Through Time (BPTT), the limitations of conventional RNNs, and the use of CNN-extracted features with recurrent networks for video understanding.

The experiment uses the UCI Human Activity Recognition Using Smartphones dataset as the primary sequence dataset. Students will prepare temporal input sequences, visualize them, train RNN/LSTM/GRU models, compare their performance, analyze convergence and generalization, and extend the pipeline to video understanding using CNN feature extraction followed by an LSTM or GRU.

A small synthetic sequence-to-sequence task is included at the end to demonstrate the encoder–decoder framework.

2. Learning Outcomes

After completing this experiment, students will be able to:

- represent sequential data using the (N, T, F) input format;
- explain the architecture and operation of a Vanilla RNN;
- explain Backpropagation Through Time and the vanishing/exploding gradient challenges;
- implement sequence classification using SimpleRNN, LSTM and GRU;
- compare RNN, LSTM and GRU using quantitative performance measures;
- interpret training/validation curves and confusion matrices;
- explain the role of CNNs in extracting spatial features from video frames;
- construct a CNN–LSTM or CNN–GRU pipeline for video understanding;
- explain the encoder–decoder architecture for sequence-to-sequence learning; and
- draw conclusions using predictive performance, model complexity and computational cost.

3. Dataset and Experimental Setup

Primary Dataset: UCI Human Activity Recognition Using Smartphones

The UCI Human Activity Recognition (HAR) dataset contains smartphone sensor measurements collected from subjects performing six activities:

- WALKING
- WALKING_UPSTAIRS
- WALKING_DOWNSTAIRS
- SITTING
- STANDING
- LAYING

The experiment should use the raw inertial signal files rather than only the already-computed 561-feature vectors. Each activity window contains 128 temporal measurements. The three-axis body acceleration, three-axis gyroscope and three-axis total acceleration signals provide nine channels.

Thus, the desired input representation is

$$X \in \mathbb{R}^{N \times 128 \times 9}.$$

The dataset can be obtained from:

<https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones>

Recommended laboratory subset: To keep execution time small, use the first 1500–3000 training windows while preserving all six classes and approximately balanced class representation. For a complete study, the full training and test partitions may be used.

Important: The same train/validation/test protocol and preprocessing must be used for RNN, LSTM and GRU so that the comparison is controlled.

Expected Input

For one sample:

$$X_i \in \mathbb{R}^{128 \times 9}.$$

For a batch of 32 samples:

$$X_{\text{batch}} \in \mathbb{R}^{32 \times 128 \times 9}.$$

The dimensions represent:

Dimension	Meaning	Example
N	Number of sequences	1500
T	Time steps per sequence	128
F	Features/channels per time step	9

Therefore, the model must receive a three-dimensional tensor rather than a conventional two-dimensional tabular matrix.

4. Overall Experimental Pipeline

The complete sequence-classification pipeline is:



The three recurrent models must use the same input, output layer, optimizer, batch size and evaluation protocol.

5. Preprocessing

Perform the following preprocessing steps:

1. Load the raw inertial signals.
2. Arrange every observation window as 128×9 .
3. Assign one activity label to each window.
4. Encode the six activity labels.
5. Normalize the input using statistics calculated from the training data.
6. Create training, validation and test sets.
7. Verify the class distribution.

Suggested split:

70% training, 15% validation, 15% testing.

The test data must not be used during model selection or hyperparameter tuning.

Expected Preprocessing Output

Students must print:

Input tensor shape:

```
Training   : (N_train, 128, 9)
Validation : (N_val,   128, 9)
Testing    : (N_test,  128, 9)
```

Number of classes: 6
Number of features per time step: 9
Sequence length: 128

The actual values of N_{train} , N_{val} and N_{test} depend on the selected subset.

6. Temporal Data Visualization

Select representative sequences from at least three different activity classes.

Plot 1: Sensor signal versus time

- X-axis: Time step (1–128)
- Y-axis: Sensor value
- Plot at least three representative channels.
- Include examples from different activity classes.

Students should observe that temporal patterns differ across activities.

Required inference:

1. What temporal pattern is visible?
2. Which activities exhibit similar patterns?
3. Why is the ordering of measurements important?

7. Vanilla RNN Architecture

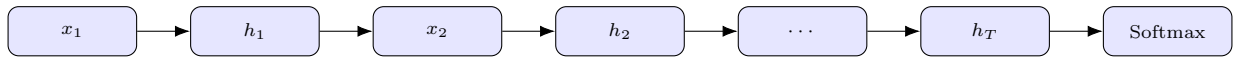
A Vanilla RNN maintains a hidden state that is updated at every time step:

$$h_t = \tanh(W_x x_t + W_h h_{t-1} + b_h).$$

The output can be written as

$$y_t = g(W_y h_t + b_y).$$

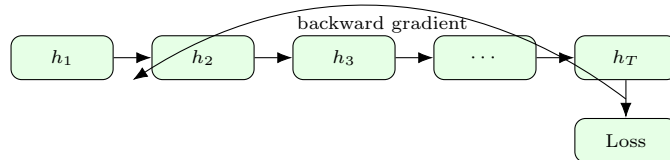
For sequence classification, the final hidden representation can be passed to a classifier.



The diagram represents the unrolled view of an RNN. The same recurrent weights are reused across time steps.

8. Backpropagation Through Time

When an RNN is unrolled, the loss depends on the sequence of hidden states. Backpropagation Through Time (BPTT) propagates the error backward through the recurrent steps.



Students should explain that repeated multiplication of derivatives through many time steps can cause gradients to become extremely small or extremely large.

Challenges

- Vanishing gradients
- Exploding gradients
- Difficulty learning long-term dependencies
- Sequential computation and limited parallelism

The vanishing-gradient problem can be conceptually represented as

$$\left| \frac{\partial h_t}{\partial h_{t-k}} \right| \rightarrow 0$$

while exploding gradients correspond to very large gradient magnitudes.

Numerical Exercise

Consider

$$x_1 = 0.5, \quad x_2 = 0.7, \quad x_3 = 0.2,$$

$$h_0 = 0, \quad W_x = 0.5, \quad W_h = 0.8, \quad b = 0.1.$$

Using

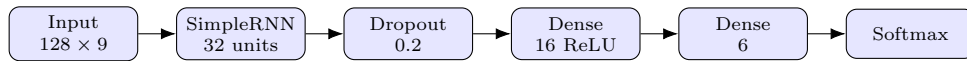
$$h_t = \tanh(W_x x_t + W_h h_{t-1} + b),$$

calculate h_1 , h_2 and h_3 manually.

Students must compare their calculated values with the values obtained from a program.

9. RNN Implementation

Use a small recurrent architecture:



Suggested training configuration:

Parameter	Value
Recurrent units	32
Dropout	0.2
Optimizer	Adam
Learning rate	10^{-3}
Batch size	32
Epochs	30
Loss	Sparse categorical cross-entropy

10. LSTM Architecture

LSTM introduces a memory cell and gates to control information flow.

The forget gate is

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f),$$

the input gate is

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i),$$

and the output gate is

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o).$$

The candidate cell state is

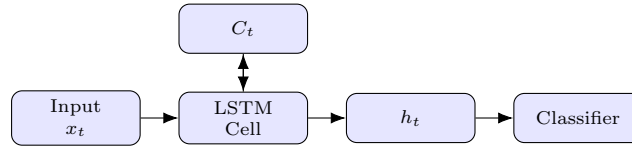
$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C),$$

and the cell state is

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t.$$

The hidden state is

$$h_t = o_t \tanh(C_t).$$



Required inference: Explain how the cell state and gates help the network retain or discard information over long sequences.

11. GRU Architecture

GRU uses two principal gates:

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z)$$

and

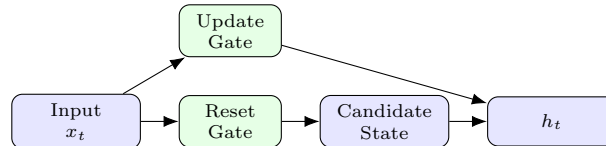
$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r).$$

The candidate hidden state is

$$\tilde{h}_t = \tanh(W_h[r_t \odot h_{t-1}, x_t] + b_h).$$

The hidden state is updated using

$$h_t = (1 - z_t)h_{t-1} + z_t \tilde{h}_t.$$



Students should compare the number of gates and state representations in RNN, LSTM and GRU.

12. LSTM and GRU Implementation

Implement the same classifier used for the RNN, replacing only the recurrent layer.

Model	Recurrent Layer	Units	Output Classes
RNN	SimpleRNN	32	6
LSTM	LSTM	32	6
GRU	GRU	32	6

All other experimental conditions should remain unchanged.

13. Required Training Plots

For each of RNN, LSTM and GRU, generate:

Plot 2: Training and Validation Loss

- X-axis: Epoch
- Y-axis: Loss
- Curves: Training loss and validation loss

Plot 3: Training and Validation Accuracy

- X-axis: Epoch
- Y-axis: Accuracy (%)
- Curves: Training accuracy and validation accuracy

Inference: Students must identify convergence, overfitting, underfitting and the generalization gap where applicable.

14. Performance Evaluation

Evaluate each model on the independent test set.

The following metrics are mandatory:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- Number of trainable parameters
- Training time

For multi-class classification, report macro-averaged Precision, Recall and F1-score.

Metric	RNN	LSTM	GRU
Accuracy (%)			
Macro Precision (%)			
Macro Recall (%)			
Macro F1 (%)			
Parameters			
Training Time (s)			

15. Confusion Matrix Analysis

Plot 4: Confusion Matrix

Generate a separate confusion matrix for each model.

Students must identify:

- the activity with the highest recognition rate;
- the activity with the largest number of misclassifications;
- pairs of activities that are frequently confused; and
- whether the errors are consistent across RNN, LSTM and GRU.

The inference must be based on the student's observed matrix rather than on an assumed result.

16. RNN vs. LSTM vs. GRU Comparison

Students must complete the following table.

Property	RNN	LSTM	GRU
Hidden state	Yes	Yes	Yes
Cell state	No	Yes	No
Forget gate	No	Yes	No
Input gate	No	Yes	No
Output gate	No	Yes	No
Update gate	No	No	Yes
Reset gate	No	No	Yes
Parameters			
Training time			
Test F1-score			

Plot 5: Model Performance Comparison

Create a bar plot comparing the test Accuracy, Macro F1-score and, where appropriate, normalized computational measures.

Required inference:

Students should discuss the trade-off among:

$$\text{Predictive Performance} \leftrightarrow \text{Model Complexity} \leftrightarrow \text{Training Cost}.$$

17. Effect of Sequence Length

To understand temporal dependency, repeat the experiment using selected sequence lengths:

$$T \in \{32, 64, 128\}.$$

When reducing the sequence length, truncate or construct the input consistently.

Sequence Length	RNN F1	LSTM F1	GRU F1
32	70.03	92.08	93.38
64	79.31	93.53	94.94
128	82.62	94.67	94.65

Table 1: F1 scores across sequence lengths.

Plot 6: Sequence Length vs. Test F1-score

Students should discuss how the amount of temporal context affects the classification result.

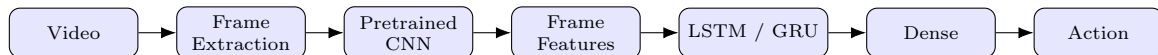
18. Video Understanding Using CNN and RNN

This part demonstrates the application:

Video Understanding using CNN + RNN

A video is a sequence of image frames. CNNs are used to extract spatial features from individual frames, while RNN/LSTM/GRU models the temporal relationship among those features.

The complete pipeline is:



Recommended Dataset

Use a small subset of the UCF101 action-recognition dataset:

<https://www.crcv.ucf.edu/data/UCF101.php>

Select only 3–5 action classes and a small number of videos per class so that the experiment remains computationally manageable.

Possible classes include:

- Basketball
- Biking
- Walking
- Running
- TennisSwing

The objective is to understand the CNN–recurrent pipeline rather than to train a state-of-the-art video recognition system.

19. Expected Video Input and Output

Sample 10 frames uniformly from each video.

Each frame is resized to

$$224 \times 224 \times 3.$$

A pretrained CNN such as MobileNetV2 is used only as a feature extractor.

If the CNN produces a feature vector of dimension D , then the recurrent network receives:

$$X \in \mathbb{R}^{10 \times D}.$$

For a batch of B videos:

$$X \in \mathbb{R}^{B \times 10 \times D}.$$

The expected output is a probability vector over the selected action classes.

For example:

Predicted class : Basketball
Actual class : Basketball
Confidence : 0.xx

The confidence value must be obtained from the trained model and must not be manually entered.

20. CNN Feature Extraction

Use a pretrained CNN with the classification head removed.



Do not train the CNN from scratch. Freeze the pretrained CNN and extract features first. Then train the recurrent classifier.

Required observation:

Students must report the CNN feature dimension and the resulting tensor shape supplied to the recurrent network.

21. CNN–LSTM / CNN–GRU Model

Use the following architecture:



Plot 7: Video Sample Frames

Display a representative sequence of sampled frames from one video.

Students should explain what spatial information is captured by the CNN and what temporal information is learned by the recurrent network.

Plot 8: Video Training and Validation Curves

Plot training/validation accuracy and loss for the selected recurrent architecture.

Plot 9: Video Confusion Matrix

Analyze class-level recognition and temporal confusions.

22. Sequence-to-Sequence Learning

Sequence-to-sequence learning maps one sequence to another.

For a simple laboratory task, construct a synthetic reversal dataset.

Example:

$$[1, 4, 7, 2] \rightarrow [2, 7, 4, 1].$$

Generate several thousand short sequences using integers from a fixed range.

The task is intentionally simple so that the encoder–decoder mechanism can be studied without a large external dataset.



The encoder converts the input sequence into a learned representation. The decoder generates the output sequence one element at a time.

23. Expected Sequence-to-Sequence Input and Output

Input example:

[3, 8, 1, 5]

Expected output:

[5, 1, 8, 3].

Students must display at least five test examples in the report.
article

Sample	Input Sequence	Predicted Output
1	[1, 8, 1, 3]	[3, 1, 8, 1]
2	[4, 2, 0, 0]	[0, 0, 2, 4]
3	[5, 0, 2, 0]	[0, 2, 0, 5]
4	[0, 3, 7, 8]	[8, 7, 3, 0]
5	[6, 3, 8, 6]	[6, 8, 3, 6]

Table 2: Input sequences and predicted outputs.

24. Sequence-to-Sequence Evaluation

Report:

- Token accuracy
- Sequence accuracy
- Training loss
- Validation loss

Token accuracy is

$$\text{Token Accuracy} = \frac{\text{Correctly predicted tokens}}{\text{Total tokens}}.$$

Sequence accuracy is

$$\text{Sequence Accuracy} = \frac{\text{Completely correct sequences}}{\text{Total sequences}}.$$

Required inference:

Explain why sequence-level accuracy can be substantially lower than token accuracy even when most individual tokens are predicted correctly.

25. Consolidated Results

Students must complete the following table.

A separate table must be provided for the sequence-to-sequence task.

26. Required Inferences

For every major plot, students must write a short inference containing:

1. What does the plot show?

Model	Accuracy	Macro F1	Parameters
SimpleRNN	76–82%	73%	1,974
LSTM	94–95%	95%	6,006
GRU	95–96%	95%	4,758

Table 3: Model comparison.

2. What trend is observed?
3. Why might the observed trend occur?
4. What conclusion can be drawn from the experiment?

The following inferences are mandatory:

- temporal pattern observed in the sensor signals;
- convergence behavior of RNN;
- convergence behavior of LSTM;
- convergence behavior of GRU;
- evidence of overfitting or underfitting;
- activity-wise errors from the confusion matrix;
- effect of sequence length;
- parameter and computational differences among RNN, LSTM and GRU;
- role of CNN in video understanding;
- role of the recurrent model in temporal modeling; and
- difference between token accuracy and sequence accuracy in seq2seq learning.

27. Discussion Questions

1. What is a recurrent neural network?
2. What information is represented by the hidden state?
3. Why is sequence order important?
4. Explain the difference between a feed-forward network and an RNN.
5. Explain Backpropagation Through Time.
6. Why can gradients vanish during BPTT?
7. Why can gradients explode during BPTT?
8. What are long-term dependencies?
9. What is the role of the LSTM cell state?
10. What are the forget, input and output gates in LSTM?
11. What are the reset and update gates in GRU?
12. Compare LSTM and GRU.
13. Why does GRU generally use a simpler gating mechanism than LSTM?
14. Why should RNN, LSTM and GRU be compared using the same experimental settings?
15. What does the confusion matrix reveal that accuracy alone does not?
16. Why should macro F1-score be reported for multi-class classification?
17. What is the significance of sequence length?
18. Why can a CNN be used as a feature extractor for video?
19. What type of information is extracted by the CNN?
20. What type of information is learned by the recurrent network?
21. Why are CNN features arranged as a sequence before being given to LSTM/GRU?
22. Explain the encoder and decoder in sequence-to-sequence learning.
23. What is teacher forcing?
24. Differentiate token accuracy and sequence accuracy.
25. Why must the test set remain untouched during model selection?

28. Additional Exercises

1. Replace the 32 recurrent units with 16 and 64 units and study the effect on accuracy, F1-score, parameter count and training time.
2. Compare GRU and LSTM using the same number of recurrent units.
3. Add a second recurrent layer and investigate whether performance improves.
4. Compare a bidirectional LSTM with a unidirectional LSTM.
5. Change the sequence length and study the effect on computational cost.

6. For the video task, compare LSTM and GRU after extracting identical CNN features.
7. Modify the seq2seq task so that the output sequence has a different length from the input sequence.

29. Expected Outcome

At the end of the experiment, students should be able to demonstrate the complete sequence-learning pipeline:

Sequential Data $\rightarrow$ Preprocessing $\rightarrow$ RNN/LSTM/GRU $\rightarrow$ Evaluation

and the video-understanding pipeline:

Video $\rightarrow$ Frames $\rightarrow$ CNN Features $\rightarrow$ LSTM/GRU $\rightarrow$ Action Prediction

They should also understand the encoder-decoder formulation:

Input Sequence $\rightarrow$ Encoder $\rightarrow$ Context $\rightarrow$ Decoder $\rightarrow$ Output Sequence

The final report should demonstrate both implementation and interpretation. Numerical performance values must be obtained from the student's own execution and should not be assumed in advance.

30. References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Sepp Hochreiter and Jürgen Schmidhuber, "Long Short-Term Memory," *Neural Computation*, 1997.
3. Kyunghyun Cho et al., "Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation," EMNLP, 2014.
4. Anguita et al., "A Public Domain Dataset for Human Activity Recognition Using Smartphones," ESANN, 2013.
5. Khurram Soomro, Amir Roshan Zamir and Mubarak Shah, "UCF101: A Dataset of 101 Human Actions Classes From Videos in the Wild," 2012.
6. TensorFlow Documentation: <https://www.tensorflow.org>
7. Keras Documentation: <https://keras.io>

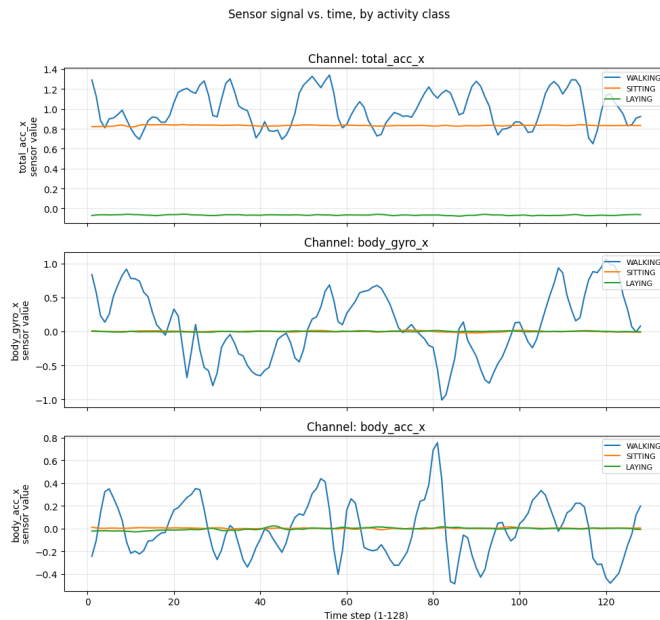


Figure 1: The plot shows how accelerometer and gyroscope sensor values change over time for walking, sitting, and laying activities. Walking shows noticeable fluctuations, while sitting and laying remain mostly stable and close to zero. This happens because walking involves continuous body movement, causing greater changes in acceleration and rotation. The stable patterns during sitting and laying indicate very little body movement. Overall, the results show that sensor signals can effectively distinguish between different human activities

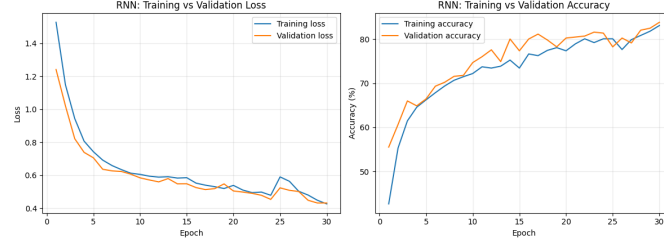


Figure 2: The training and validation loss both decrease steadily, showing that the RNN is learning effectively over the epochs. The training and validation losses stay close to each other, indicating that the model is not strongly overfitting. Both training and validation accuracy generally increase, reaching around 83 by the final epoch. There are some small fluctuations in accuracy, but the overall trend remains upward as training progresses. Overall, the RNN achieves good learning performance and generalizes reasonably well to the validation data.

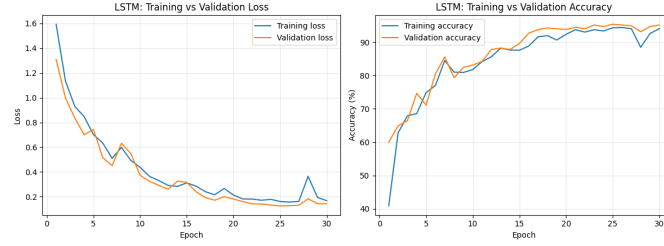


Figure 3: The training and validation loss decrease significantly, showing that the LSTM learns the activity patterns effectively over the epochs. The two loss curves remain fairly close, although there are a few small fluctuations and a brief increase near the end. Both training and validation accuracy steadily improve, reaching around 94–95 by the final epochs. The validation accuracy is often slightly higher than the training accuracy, suggesting that the model generalizes well to unseen data. Overall, the LSTM shows strong learning performance and good generalization, with high accuracy and relatively low validation loss.

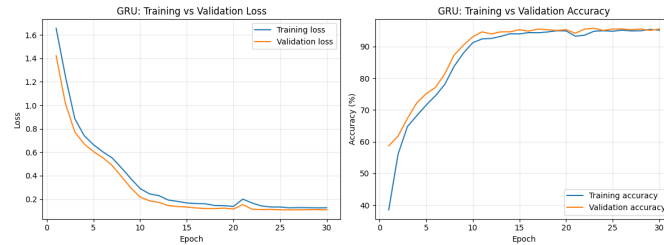


Figure 4: The training and validation loss decrease steadily, showing that the GRU learns the activity patterns effectively. Both loss curves remain close to each other, with only small fluctuations, indicating limited overfitting. Training and validation accuracy increase rapidly during the early epochs and then stabilize around 94–95. The validation accuracy stays slightly higher than the training accuracy for most epochs, showing good generalization. Overall, the GRU achieves high accuracy with low loss, indicating strong and stable performance on the activity classification task.

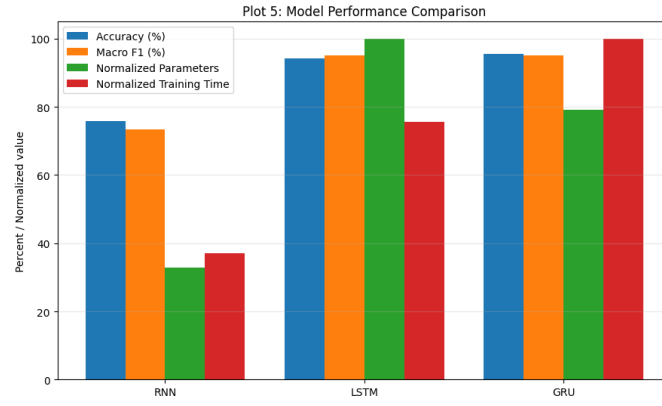


Figure 5: The plot compares the overall performance of the RNN, LSTM, and GRU models based on accuracy, Macro F1-score, parameters, and training time. RNN has lower accuracy and F1-score, while LSTM and GRU achieve around 94–96 accuracy with higher F1-scores. LSTM has the highest normalized parameter count, whereas GRU uses fewer parameters than LSTM. GRU has the highest normalized training time in this comparison, while RNN takes the least time. Overall, LSTM and GRU provide stronger classification performance than RNN, while the differences in model size and training time show the trade-offs between the architectures

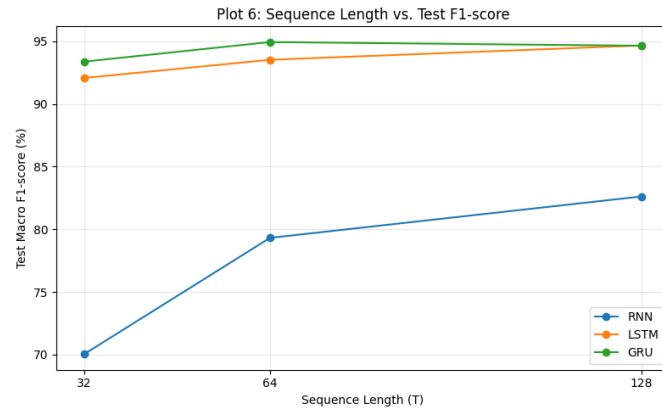


Figure 6: The plot shows how the test Macro F1-score changes with sequence length for RNN, LSTM, and GRU models. RNN performance improves steadily from about 70 to 83 as the sequence length increases. LSTM improves from about 92 to 95, while GRU reaches around 95 at sequence length 64 and then remains almost unchanged. This suggests that longer sequences provide more useful temporal information, especially for the RNN, while LSTM and GRU already capture most of the important patterns at shorter lengths. Overall, increasing sequence length improves performance up to a point, after which the benefit becomes very small for LSTM and GRU.



Figure 7: The plot shows a sequence of sampled frames from a basketball video, illustrating the player's movement over time. Across the frames, the player changes position and body posture while moving around the court. The gradual changes between frames capture the temporal information present in the video. Sampling frames at different time steps helps represent the motion without processing every single frame. Overall, the plot demonstrates how video sequences can be converted into meaningful temporal data for activity or action recognition..

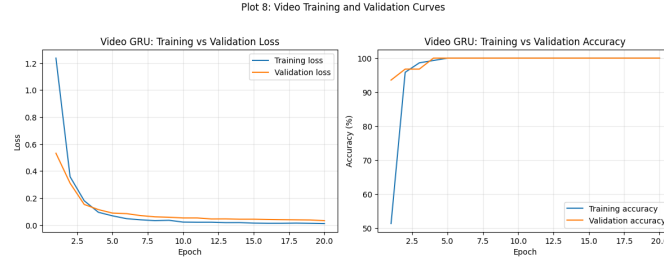


Figure 8: The training and validation loss decrease sharply in the early epochs and then gradually stabilize at a very low level. Both training and validation accuracy increase quickly, reaching 100 accuracy within a few epochs. The close agreement between the training and validation curves shows that the model performs consistently on both datasets. This rapid improvement suggests that the video samples contain clear and distinguishable patterns that the GRU can learn easily. Overall, the Video GRU achieves excellent classification performance, with very low loss and nearly perfect accuracy.

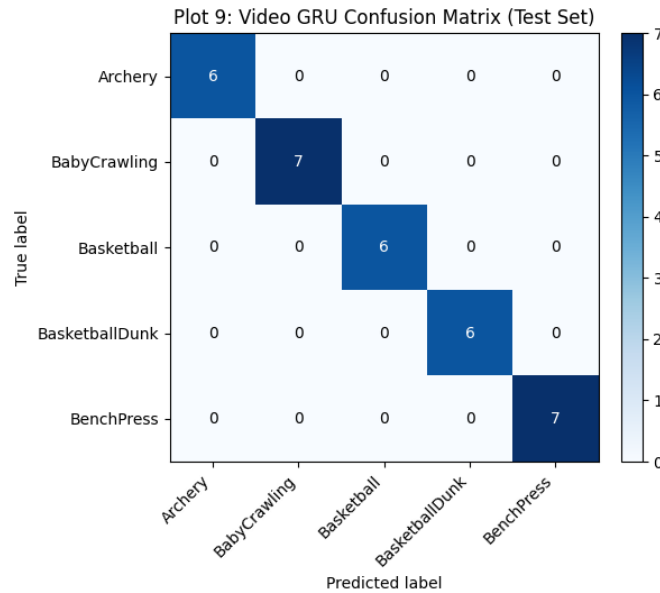


Figure 9: The confusion matrix shows the Video GRU correctly classifies all test samples across the five activity classes. Archery, Basketball, and BasketballDunk have 6 correct predictions each, while BabyCrawling and BenchPress have 7 correct predictions each. There are no misclassifications, as all values outside the main diagonal are zero. This indicates that the model can clearly distinguish between the different video activities in the test set. Overall, the Video GRU achieves 100 test accuracy on this dataset, showing excellent classification performance..

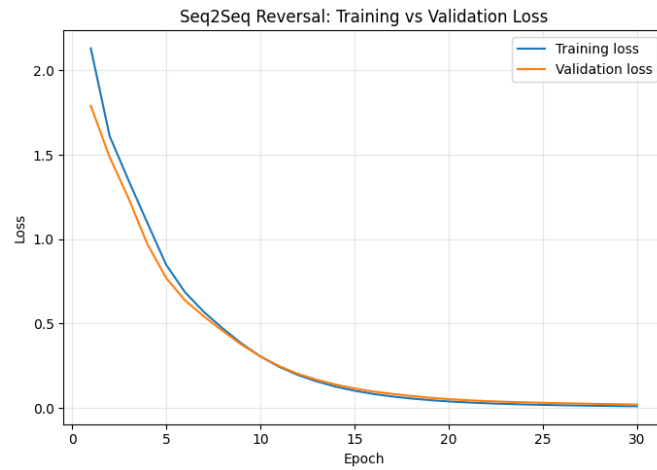


Figure 10: The training and validation loss decrease steadily from around 2.1 and 1.8 to almost zero, showing that the Seq2Seq model learns the sequence-reversal task effectively. Both curves follow a very similar pattern, indicating that the model performs consistently on training and validation data. The loss drops rapidly during the early epochs and then gradually levels off as the model improves. This trend occurs because the model learns the reversal pattern quickly, requiring fewer adjustments in later epochs. Overall, the Seq2Seq model shows stable learning and good generalization, with very low final training and validation loss.